

Chapter-2: Laplace Transform

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1. $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$ (Definition of Laplace transform)

Example-1:

- (a) $\mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0$
- (b) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a$
- (c) $\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}, \quad s > -1$
- (d) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 1, 2, \dots$

2. $\mathcal{L}\{f(t)\}$ exists if $|f(t)| \leq Me^{ct}$, for some constants M , c , and $T > 0$.

Example-2:

- (a) $\mathcal{L}\left\{\frac{1}{t}\right\}$ does not exist.
- (b) $\mathcal{L}\left\{e^{t^2}\right\}$ does not exist

3. $\mathcal{L}^{-1}\{F(s)\} = f(t) \Leftrightarrow F(s) = \mathcal{L}\{f(t)\}$

Example-3:

- (a) $\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$
- (b) $\mathcal{L}^{-1}\left\{\frac{1}{s-7}\right\} = e^{7t}$
- (c) $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$

4. $\mathcal{L}$ and $\mathcal{L}^{-1}$ are linear operators.

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

$$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$$

Example-4:

- (a) $\mathcal{L}\{6 - t^7 + 9e^{-2t} + 3\cos t\} = \frac{6}{s} - \frac{7!}{s^8} + \frac{9}{s+2} + \frac{3s}{s^2+1}$
- (b) $\mathcal{L}^{-1}\left\{\frac{3}{2s+7} - \frac{8}{s^2+3} + \frac{4}{s^9}\right\} = \frac{3}{2}e^{\frac{-7}{2}t} - \frac{8}{\sqrt{3}}\sin\sqrt{3}t + \frac{4}{8!}t^{8t}$

5. Not every function of s is a Laplace transform of a piecewise continuous function, $f(t)$, of exponential order.

- If $\mathcal{L}\{f(t)\}$ exists, then
 - $f(t)$ is piecewise continuous and exponential order.
 - $\lim_{s \rightarrow \infty} \mathcal{L}\{f(t)\} = 0$

Example-5:

$F(s) = s^2$ is not the Laplace transform of a piecewise continuous function of exponential order because $\lim_{s \rightarrow \infty} s^2 = \infty$

6. Laplace and Inverse Laplace Transforms of Basic Functions:

- (a) $\mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0 \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$
- (b) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
- (c) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 1, 2, \dots, \quad s > 0 \quad \text{and}$
 $\mathcal{L}^{-1}\left\{\frac{c}{s^m}\right\} = \frac{c}{(m-1)!} t^{m-1}, \quad m = 1, 2, \dots$
- (d) $\mathcal{L}\{\cos bt\} = \frac{s}{s^2+b^2}, \quad s > 0 \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{s}{s^2+b^2}\right\} = \cos bt$
- (e) $\mathcal{L}\{\sin bt\} = \frac{b}{s^2+b^2}, \quad s > 0 \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{b}{s^2+b^2}\right\} = \sin bt$
- (f) $\mathcal{L}\{\cosh bt\} = \frac{s}{s^2-b^2}, \quad s > b \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{s}{s^2-b^2}\right\} = \cosh bt$
- (g) $\mathcal{L}\{\sinh bt\} = \frac{b}{s^2-b^2}, \quad s > b \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{b}{s^2-b^2}\right\} = \sinh bt$

7. Partial fractions and completing square are very important to evaluate inverse Laplace transform:

Example-7: Evaluate $\mathcal{L}^{-1}\left\{\frac{3s-2}{s^3(s^2+4)}\right\}$

Solution: Assume $\frac{3s-2}{s^3(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{Ds+E}{s^2+4}$

$$3s-2 = As^2(s^2+4) + Bs(s^2+4) + C(s^2+4) + (Ds+E)s^3$$

- Setting $s = 0$, gives $C = -\frac{1}{2}$.
- Comparing coefficients of s^4, s^3, s^2 , and s , we get

$$\begin{aligned} 0 &= A + D & (1) \\ 0 &= B + E & (2) \\ 0 &= 4A + C & (3) \\ 3 &= 4B & (4) \end{aligned}$$

From (3), $A = \frac{1}{8}$, from (4), $B = \frac{3}{4}$, from (1) $D = -\frac{1}{8}$ and from (2), $E = -\frac{3}{4}$

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{3s-2}{s^3(s^2+4)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1/8}{s} + \frac{3/4}{s^2} + \frac{-1/2}{s^3} + \frac{-s/8 - 3/4}{s^2+4}\right\} \\ &= \frac{1}{8} + \frac{3}{4}t - \frac{1}{4}t^2 - \frac{1}{8}\cos 2t - \frac{3}{8}\sin 2t \end{aligned}$$

8. First-Shifting (translation) Theorem:

- For any real number a and $F(s) = \mathcal{L}\{f(t)\}$,
$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \mathcal{L}\{f(t)\}|_{s \rightarrow s-a} \\ &= F(s)|_{s \rightarrow s-a} \\ &= F(s-a)\end{aligned}$$
- and
- $$\begin{aligned}\mathcal{L}^{-1}\{F(s-a)\} &= \mathcal{L}^{-1}\{F(s)|_{s-a \rightarrow s}\} \\ &= e^{at}f(t)\end{aligned}$$

Example-8: Evaluate:

- (a) $\mathcal{L}\{e^t \cos^2 3t\}$
(b) $\mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3} + \frac{2s+5}{s^2+6s+34}\right\}$

Solution: (a) Let $f(t) = e^t \cos^2 3t = \frac{1}{2}(1 + \cos 6t)e^t$. **Then**

$$\begin{aligned}\mathcal{L}\{e^t \cos^2 3t\} &= \mathcal{L}\left\{\frac{1}{2}(1 + \cos 6t)e^t\right\} \\ &= \mathcal{L}\left\{\frac{1}{2}(1 + \cos 6t)\right\}\Big|_{s \rightarrow s-1} \\ &= \frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2+36}\right)\Big|_{s \rightarrow s-1} \\ &= \frac{1}{2}\left(\frac{1}{s-1} + \frac{s-1}{(s-1)^2+36}\right)\end{aligned}$$

Solution: (b) Let $F(s) = \frac{(s+2)^2}{s^3} + \frac{2s+5}{s^2+6s+34} = \frac{1}{s} + \frac{4}{s^2} + \frac{4}{s^3} + \frac{2s+5}{(s+3)^2 25}$

$$\begin{aligned}F(s) &= \frac{1}{s} + \frac{4}{s^2} + \frac{4}{s^3} + \frac{2(s+3-3)+5}{(s+3)^2 25} \quad (\text{Adding zero}) \\ &= \frac{1}{s} + \frac{4}{s^2} + \frac{4}{s^3} + \frac{2(s+3)}{(s+3)^2 25} + \frac{-1}{(s+3)^2 + 25} \\ &= \frac{1}{s} + \frac{4}{s^2} + \frac{4}{s^3} + \frac{2(s+3)}{(s+3)^2 25} - \frac{1}{5} \frac{5}{(s+3)^2 + 25} \quad (\text{Multiplying by one.}) \\ \therefore, \mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3} + \frac{2s+5}{s^2+6s+34}\right\} &= 1 + 4t + 2t^2 + 2e^{-3t} \cos 5t - \frac{1}{5}e^{-3t} \sin 5t.\end{aligned}$$

Steps to Evaluate $\mathcal{L}\{e^{at}f(t)\}$ and $\mathcal{L}^{-1}\{F(s-a)\}$ using FST:

- Identify: a and $f(t)$; a and $F(s)$.
- Evaluate: $\mathcal{L}\{f(t)\}$; $\mathcal{L}^{-1}\{F(s)\}$.
- Use FST: Replace each s by $s-a$; Multiply result by e^{at}

9. Second-Shifting (Translation) Theorem (SST or STT):

- For any real number $a \geq 0$, the function $u(t-a)$ or $H(t-a)$ defined by

$$u(t-a) = H(t-a) = \begin{cases} 0, & 0 \leq t < a, \\ 1, & t \geq a \end{cases}$$

is called the **unit-step (Heaviside)** function.

- For any piecewise continuous function $f(t)$,

$$f(t)H(t-a) = \begin{cases} 0, & 0 \leq t < a, \\ f(t), & t \geq a \end{cases}$$

- Any piecewise continuous function of the form:

$$f(t) = \begin{cases} f_1(x), & 0 \leq t < a, \\ f_2(t), & t \geq a \end{cases}$$

can be written in the form

$$f(t) = f_1(t) + [f_2(t) - f_1(t)] H(t-a)$$

For example, if

$$f(t) = \begin{cases} t^2, & 0 \leq t < \pi, \\ \sin 2t, & t \geq \pi \end{cases}, \text{ then } f(t) = t^2 + (\sin 2t - t^2) H(t-\pi)$$

SST or STT: For any real number $a \geq 0$ and $F(s) = \mathcal{L}\{f(t)\}$,

- $\mathcal{L}\{f(t-a)H(t-a)\} = e^{-as}F(s).$
- $\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)..$
- $\mathcal{L}\{f(t)H(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}.$

Example-9: Evaluate

(a) $\mathcal{L}\{t^2 + (\sin 2t - t^2) H(t-\pi)\}$

(b) $\mathcal{L}^{-1}\left\{\frac{se^{-s}}{s^2-6s+10}\right\}$

Solution: (a) $\mathcal{L}\{t^2 + (\sin 2t - t^2) H(t-\pi)\} = \mathcal{L}\{t^2\} + \mathcal{L}\{(\sin 2t - t^2) H(t-\pi)\}$

- $\mathcal{L}\{t^2\} = \frac{2}{s^3}.$

- $\mathcal{L}\{(\sin 2t - t^2) H(t-\pi)\} = \mathcal{L}\{\sin 2(t-\pi) - (t+\pi)^2\}$
 $= \mathcal{L}\{\sin 2t - t^2 - 2\pi t - \pi^2\}$
 $= \frac{2}{s^2+4} - \frac{2}{s^3} - \frac{2\pi}{s^2} - \frac{\pi^2}{s}$

$$\therefore, \mathcal{L}\{t^2 + (\sin 2t - t^2) H(t-\pi)\} = \frac{2}{s^3} + e^{-\pi s} \left(\frac{2}{s^2+4} - \frac{2}{s^3} - \frac{2\pi}{s^2} - \frac{\pi^2}{s} \right)$$

Solution: (b) Let $F(s) = \frac{s}{s^2-6s+10} = \frac{s-3+3}{(s-3)^2+1} = \frac{s-3}{(s-3)^2+1} + \frac{3}{(s-3)^2+1}$

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = e^{3t} \cos t + 3e^{3t} \sin t$$

$$\mathcal{L}^{-1}\left\{\frac{se^{-s}}{s^2-6s+10}\right\} = \left[e^{3(t-1)} \cos(t-1) + 3e^{3(t-1)} \sin(t-1) \right] H(t-1)$$

10. **Derivatives of a Transform:**

$$\begin{aligned}\mathcal{L}\{t^n f(t)\} &= (-1)^n \frac{d^n}{ds^n} \mathcal{L}\{f(t)\} \\ &= (-1)^n F^{(n)}(s),\end{aligned}$$

where $F(s) = \mathcal{L}\{f(t)\}$ and $n = 1, 2, \dots$

Example-10: Evaluate $\mathcal{L}\{te^{-3t} \cos 2t\}$

Solution: $\mathcal{L}\{te^{-3t} \cos 2t\} = -\frac{d}{ds} \mathcal{L}\{e^{-3t} \cos 2t\} = -\frac{d}{ds} \frac{s+3}{(s+3)^2+4} = \frac{s^2+6s-4}{(s^2+6s+13)^2}$

11. **Transformations of Derivatives:**

- $\mathcal{L}\{y'(t)\} = s\mathcal{L}\{y(t)\} - y(0)$
 $= sY(s) - y(0)$
- $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$

Example-11: Solve the initial-value problems

(a) $y'' - 6y' + 9y = t^2 e^{3t}$, $y(0) = 2$, $y'(0) = 6$

(b) $y'' + 16y = \cos 4t$, $y(0) = 0$, $y'(0) = 1$

Solution: (a) $\mathcal{L}\{y'' - 6y' + 9y\} = \mathcal{L}\{t^2 e^{3t}\}$

$$s^2 Y(s) - sy(0) - y'(0) - 6(sY(s) - y(0)) + 9Y(s) = \frac{2}{(s-3)^3}$$

$$Y(s)(s-3)^2 = 2s - 6 + \frac{2}{(s-3)^3}$$

$$Y(s) = \frac{2}{s-3} + \frac{2}{(s-3)^5}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1} \left\{ \frac{2}{s-3} + \frac{2}{4!} \frac{4!}{(s-3)^5} \right\}$$

$$y(t) = \frac{1}{4} \sin 4t + \frac{1}{8} t \cos 4t \text{ is the solution of the initial - value problem.}$$

Solution: (b) $\mathcal{L}\{y'' - 16y\} = \mathcal{L}\{\cos 4t\}$

$$s^2 Y(s) - sy(0) - y'(0) - 16Y(s) = \frac{s}{s^2 + 16}$$

$$Y(s)(s^2 - 16) = 1 + \frac{s}{s^2 + 16}$$

$$Y(s) = \frac{1}{s^2 + 16} + \frac{s}{(s^2 + 16)^2}$$

$$\Rightarrow y(t) = \frac{1}{4} \mathcal{L}^{-1} \left\{ \frac{4}{s^2 + 16} \right\} + \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{8s}{(s^2 + 16)^2} \right\}$$

$$y(t) = \frac{1}{4} \sin 4t + \frac{1}{12} t^4 e^{3t} \text{ is the solution of the initial - value problem.}$$

12. Transforms Integrals:

$$\begin{aligned}\mathcal{L}\left\{\int_0^t f(u)du\right\} &= \frac{1}{s}\mathcal{L}\{f(t)\} \\ &= \frac{1}{s}F(s),\end{aligned}$$

where $F(s) = \mathcal{L}\{f(t)\}$